4. Data Generation and Preparation

Chapter 3 defined two plant representations: detailed hydraulics in Simscape and fast rigid-body dynamics in IsaacLab. This chapter documents the datasets that connect them including collection, auditing, feature construction and splitting. Table 3.3 records their provenance.

Simscape campaigns. Closed-loop runs provide realistic operating data for the inverse and reduced-order models. Separate open-loop valve excitation provides the causal command information needed by the strict forward model (Section 4.4).

ROM calibration. After Simscape training, the command ROM was fine-tuned on five closed-loop IsaacLab reference shapes. Only its bounded correction head remained trainable, with Simscape replay retained (Section 4.6). The ID–FD pair was not calibrated in IsaacLab.

IsaacLab adaptation data. The system-dynamics surrogate, privileged encoder and AAM use a separate 118-run collection from the deployed command ROM–PhysX plant, including dedicated PRBS windows (Section 4.7). No earlier adaptation or system-dynamics model was reused. Figure 4.1 summarizes the common preparation workflow.